Q1

Calculate the intrinsic carrier density n_i for silicon at $T = 50$ K and 350 K.

Ans. $9.6 \times 10^{-39}/\text{cm}^3$; $4.15 \times 10^{11}/\text{cm}^3$

Q2

For a silicon crystal doped with boron, what must N_A be if at $T = 300\ \text{K}$ the electron concentration drops below the intrinsic level by a factor of 10^6?

Ans. $N_A = 1.5 \times 10^{16}/\text{cm}^3$